CASCADE: stage-aware shipment ETA, delay propagation and explanation

Aiqon AI Internship take-home · model v1.1.0 · seed 42 · out-of-time test (Q4 peak) · 6/6 quality gates pass · generated 2026-10-13

> **Answer first.** On a held-out, later-in-time test season, CASCADE's ETA error (MAE of the P50 final delay) is **24.5 h**, which is **36% better than carry-forward** (95% CI 24%–48%) and 30% better than a lane-level propagation prior. Its P10–P90 ranges cover 80% of outcomes (target 80%), P(late > 24 h) has Brier 0.181 and PR-AUC 0.86, and every P50 is shown as realised delay plus a per-stage projection of the delay still to come. Because a structural causal simulator records a *cause ledger* for every hour of delay, we can score that explanation: the stage projected to add the most future delay is the ledger's top future stage 42% of the time (top-2: 69%). Caveat: the point forecast **and** these stage explanations come from a mechanistic operator-rulebook projection; the learned median correction did not beat it out-of-time, so validation set its weight to 0.0. The ML contributes the P10–P90 spread, P(late) and the stage forecasts.

1 · Problem and definitions

A container shipment moves through a fixed **stage graph**: gate-in → origin departure → (hub arrival → hub departure) → destination arrival → discharge → customs release → gate-out → delivery. Direct lanes skip the hub. Each milestone e_k has a planned and an actual time. All quantities are in hours, and positive means late.

`delay(`e_k`)`	= actual(e_k) − planned(e_k)
Δ_k	= delay(e_k) − delay(e_{k-1}): the stage increment (positive adds delay, negative absorbs it)
`D`	= delay(DELIVERED) = $\Sigma_k \Delta_k$: total delay; the model predicts the *remaining* part R(t) = D − delay_{obs}(t)
`ETA`$_q$`(t)`	= planned delivery + observed delay so far + predicted remaining delay $\hat{R}_q(t)$, for q ∈ {P10, P50, P90}
`late`	= 1[D > 24 h]; P(late) is reported with every ETA

Predictions are **as-of nowcasts**: at each of 9 checkpoints (booked … gate-out) the model sees only what was known at that instant, including *forecasts* (not realised values) of future congestion and weather.

2 · Data and ground truth

Instead of adding random noise to timestamps, we built a **structural causal model (SCM)** of 300 shipments on lanes between real Asian origins, hubs (Singapore, Colombo, Jebel Ali, Tanjung Pelepas) and European / US destinations. Shipments share voyages, so errors are correlated. Each mechanism below writes signed hours into a **cause ledger** (`shipment, stage, cause, hours`). Within a stage the ledger sums exactly to Δ_k, and over all stages to D. The true root cause is the stage with the largest positive Δ_k. Late rate in this run: 38%.

Table 1 · Logistics mechanics in the simulator (each produces ledger entries)

Stage	Real-world behaviour modelled
Origin	Carrier reliability (latent), origin congestion and storms delay departure (shared by the voyage).
Ocean	Weather slows the vessel; the vessel **recovers** part of an incoming delay by speeding up (negative Δ, capped).
Transshipment	Hub handling (longer when congested) must finish before the connection cut-off. A **missed connection waits for the next sailing** (weekly or twice-weekly headway), which is the main cascade amplifier.
Destination port	Kingman-like **berth queue** W ∝ ρ/(1−ρ); a vessel arriving late **loses its berth window** and waits longer (congestion × lateness interaction).
Customs	Inspection probability rises with commodity risk and incomplete documents; release is shifted to the next **business slot** (weekends, local holidays).
Inland	Rail leaves at fixed times, so a missed **cut-off** waits for the next train; trucks need appointments; consignee **receiving windows**.

External factors. Daily port congestion ρ(t) (base + seasonality incl. the Q4 peak and the Lunar New Year backlog, plus AR(1) noise) and regional weather with storm episodes and the monsoon. For future dates the model only sees noisy *as-of forecasts* whose error grows with the horizon. **Missingness is typed:** *structural* (no hub on direct lanes → sentinel, never NaN); *MCAR* reporting gaps (rate 6%); *MNAR* gaps from poor carrier EDI feeds (up to 25%) and congested terminals that fail to report discharge. A report reaches the system after a carrier-dependent latency (median 0.5–6 h by feed quality), and delivery is always observed. **Split:** out-of-time. The last voyages by departure, which fall in the Q4 peak, form the test set (62 shipments, 725 as-of snapshots), so the test season is shifted relative to training. **Assumptions:** fixed schedules and headways per lane; costs from a configurable tariff; no strikes or blank sailings; one container per shipment.

3 · Approach

- **As-of checkpoints.** Each shipment expands into one row per milestone. Features use only milestone reports that had *arrived* by the as-of time (an event that happened but was reported later is invisible), and leakage is property-tested.

- **Operator rulebook vs truth.** A physics-style projection of every *remaining* stage (cut-offs, headways, Kingman queue, customs calendars, rail timetables) from forecast congestion and weather, with coefficients **deliberately different from the simulator's**, so the model cannot "know the simulator".
- **Propagation features.** These include projected hub buffer versus cut-off, congestion × current lateness, days to the next business slot, rail cut-off slack, and missingness flags that carry signal (EDI tier, gaps so far).
- **Residual quantile GBM with temporal shrinkage.** LightGBM models P10/P50/P90 of the residual on top of the rulebook (300 trees, 15 leaves). The learned median correction is blended in with weight $\lambda \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$, chosen on the latest training voyages. **$\lambda = 0.0$ was selected**, so the median comes from the rulebook, and ML contributes uncertainty, P(late) and stage forecasts.
- **Drift-aware CQR and P(late).** Conformalised quantile regression per checkpoint group (offset = max(out-of-fold, temporal-slice)) protects coverage under the seasonal shift. P(late) blends a monotone classifier + isotonic calibration with the quantile-implied probability (weight from {0.0, 0.5, 1.0}; selected 0.5).
- **Explanations.** Stage bar = rulebook projected increment + λ × exact TreeSHAP of the residual model (`pred_contrib`); at $\lambda = 0$ the bars, and the attribution score, are the **rulebook alone**. Realised delay + stage bars + a floor/ordering adjustment = P50; templates write ops/customer/analyst narratives. Future-stage attribution is inherently noisy, so we report top-2 and lift, not just top-1.
- **Decision layer.** Promise ETA with hysteresis, alerts, cost-of-delay, an intervention advisor (model re-run under a fixed action menu), a scenario radar and a prediction ledger; FastAPI with a fallback chain (lane propagation prior → carry-forward) and a Streamlit control tower.

4 · Results

Table 2 · Final-delay error on the out-of-time test, all 725 snapshots (hours; bias = mean of prediction − actual)

Model	MAE	RMSE	MedAE	P90 AE	Bias
B0 Schedule (assume on plan)	56.3	88.6	48.0	168.0	-55.9
B1 Carry-forward (current delay persists)	38.5	64.5	22.5	88.7	-31.3
B2 Propagation prior (lane-level)	34.9	59.0	20.2	81.7	-21.3
Operator rulebook projection (no ML)	24.5	50.8	1.5	72.0	-16.7
CASCADE (rulebook + residual GBM, P50)	**24.5**	**50.8**	**1.6**	**72.0**	**-16.7**

MAE improvement, bootstrapped over the 20 test *voyages* (the independent unit): **36%** vs carry-forward (95% CI 24%–48%) and **30%** vs the propagation prior (CI 18%–43%). Against the **rulebook alone** CASCADE is 0.0% (CI -0.3% to 0.3%): with $\lambda = 0.0$ the gain over the baselines is the rulebook's. The negative bias reflects the Q4 shift. **Original targets not (fully) met:** G2b brier p(late > 24 h) (unmet); G3 p10-p90 coverage in every checkpoint group (partly); G4 future-delay driver top-1 vs cause ledger (unmet); G6b promise changes / raw p50 day changes (unmet) — full table in the model card.

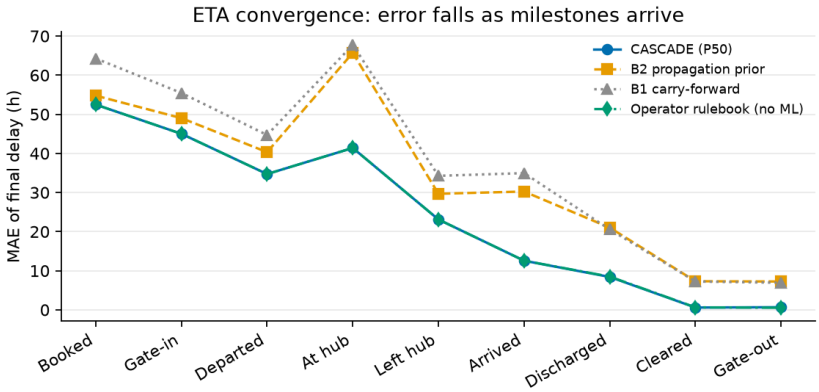


Figure 1 · ETA convergence. The error of the final-delay forecast falls as milestones arrive (MAE 52.5 h at booking → 0.7 h at gate-out). The spike at the hub is the missed-connection cliff: it is resolved only once the hub departure is known.

Table 3 · Uncertainty and P(late)

Checkpoint group	P10–P90 coverage	Mean width
Pre-departure	77%	6.5 days
At sea	78%	4.3 days
At port	81%	2.2 days
Post-customs	89%	11 h
Overall	**80%**	**3.5 days**

Table 4 · Explanation fidelity vs simulator ground truth

Root-cause stage (351 pre-arrival snapshots)	Accuracy
CASCADE stage ranking (rulebook at $\lambda = 0$), top-1 · top-2	**42% · 69%**
… top-1 on major delays only	50%
Reconciled stage forecast, top-1	43%
Lane-history explainer, top-1	26%
Random future stage	18%

P(late > 24 h)	Brier	PR-AUC	ROC-AUC
All snapshots (base rate 53%)	0.181	0.86	0.82
After destination arrival	0.144	0.91	0.88

Calibration ECE 0.108. On temporal validation the blend of classifier and quantile-implied P(late) lowered Brier from 0.213 (quantile-only) and 0.158 (classifier-only) to 0.151.

Propagation factor Φ = ΔD / Δstage	Model	True
Destination arrival	1.26	1.58
Hub arrival	1.69	2.39
Origin departure	1.67	1.90
Discharge	0.78	0.76

Φ_{true} comes from re-running the SCM with the same random state after injecting +12 h at a stage. Spearman over 190 shipment × stage pairs: 0.46. True Φ is largest at Hub arrival (2.39) (the next-sailing effect) and smallest at Discharge (0.76); Φ > 1 means the delay amplifies downstream.

Table 5 · Ablations and robustness to missing milestones

Variant	MAE	Coverage
Full model	24.5 h	80%
− propagation / projection features	34.6 h	68%
− route / port / carrier priors	24.5 h	83%
− data-quality (missingness) flags	24.6 h	79%

Extra milestones dropped	MAE	MAE B1	Coverage
0%	24.1 h	38.4 h	80%
10%	24.3 h	38.2 h	82%
30%	28.5 h	39.9 h	81%
50%	32.3 h	42.1 h	83%

Only removing propagation features hurts; under heavy missingness CASCADE stays ahead of carry-forward. **Wrong-assumption sensitivity** (6 worlds with changed simulator mechanics): gain vs carry-forward 30%–50%, CASCADE/rulebook MAE ratio 0.80–1.00, coverage 76%–87%. Where the rulebook is wrong (*hub congestion slowdown x2*) validation picks λ = 1.0: 23.3 h vs 29.2 h.

Table 6 · Stability over 5 simulator seeds (42, 7, 11, 23, 99)

Metric	Seed 42	Mean ± sd
MAE vs carry-forward	36%	28% ± 6%
MAE vs propagation prior	30%	22% ± 7%
P10–P90 coverage	80%	81% ± 3%
Brier P(late)	0.181	0.166 ± 0.018
PR-AUC P(late)	0.86	0.87 ± 0.02
Root cause top-1	42%	47% ± 4%
Lift vs lane-history	+16 pp	+9 pp ± 6 pp
Propagation Spearman	0.46	0.35 ± 0.06
Advisor Spearman	0.44	0.45 ± 0.16

Seed 42 is the most favourable of the 5 seeds for the MAE gain; the mean is the fairer estimate. Weakest claim: root-cause lift over the lane-history explainer (+9 pp ± 6 pp), positive on average but not reliably large.

Table 7 · Advisor: hours saved, model vs SCM truth

Action	Model	True
Hub priority loading	19 h	34 h
Pre-file customs docs	3 h	18 h
Priority berth	10 h	21 h
Rail → truck	28 h	27 h
Weekend customs	28 h	6 h

Advisor ranking vs truth: Spearman 0.44, direction agreement 72% (n = 131). Largest miss: Weekend customs (28 h predicted vs 6 h true). **Decision replay:** late-alert recall 100%, precision 79%, false-positive rate 26%; 8.6 alerts per 100 shipment-days; median lead 37.0 d before *actual* delivery. The promise date holds in 89% of cases, with 0.63 reversals per shipment vs 0.79 for the raw ETA. **Latency:** predict + explain p50 1.2 ms, p95 1.7 ms (in-process, sequential, 28-thread laptop CPU, LightGBM 4.7.0).

Worked example: SHP-000174 (CNSHA-MYTPP-NLRTM), as of 2025-10-10 09:32 UTC

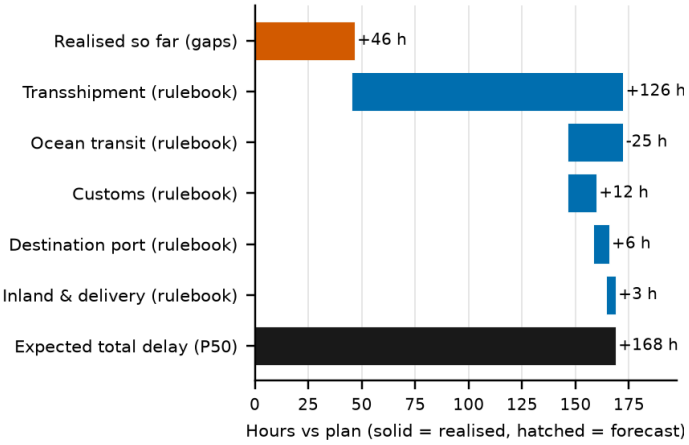


Figure 2 · Additive waterfall: realised delay (solid) + projected stage contributions (hatched) = P50 (168 h).

> SHP-000174 is expected 7.0 days late (likely 3.3 days–9.4 days late); 82% chance of more than 24 h late. Already 46 h behind plan. Main driver: Transshipment — projected to reach Tanjung Pelepas after the connection cut-off (buffer short by 36 h) (+126 h). Also contributing: Customs — customs release projected to fall outside office hours / on a non-working day at Rotterdam (+12 h). Suggested action: Check the next sailing from the hub and consider re-booking or priority loading. Confidence is reduced: 2 milestone(s) missing from the feed.

The main driver (Transshipment, 126 h) is a projected missed connection: one more sailing headway. The cause ledger agrees: the true root cause is **Transshipment**. Expected cost of delay $774 (P90 $922).

5 · Limitations and next steps

Synthetic mechanisms, not fitted to real data (schedules never revised, messages never corrected); **associational** counterfactuals (advisor Spearman 0.44 vs true interventions); **small n**: 62 test shipments in 20 voyages, so CIs are wide and resampled by voyage. **Next:** schedule revisions, a calibrated P80 promise, validation-tuned alert thresholds, an AIS/EDI adapter (`docs/ImprovementPlan.md`).